Code No. 9310

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) III-Semester (MAIN) Examinations DEC-2025/ JAN-2026

ENGINEERING MATHEMATICS –III

(Numerical Methods)

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Obtain a partial differential equation by eliminating the arbitrary function from $z = f(x^2 + y^2)$.	2	1	1
b)	Find the particular integral of $(D^2 - DD' - 20D'^2)z = e^{4x-8y}$	2	2	1
c)	Classify the partial differential equation $4\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + 5\frac{\partial^2 z}{\partial y^2} - \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$	2	1	2
d)	Using three iterations of the Newton-Raphson method, obtain approximate values of $\sqrt{31}$. Use suitable initial approximation.	2	2	3
e)	Find the second divided difference of $f(x) = \frac{1}{x}$, using the points x_0, x_1, x_2.	2	3	4
f)	Evaluate $y(0.1)$ by Taylor's series method if $y(x)$ satisfies $y' = xy + 1,\ y(0) = 1$.	2	2	5
2. a)	Obtain the general solution of Lagrange's equation $2yzp + zxq = 3xy$.	6	3	1
b)	Solve $2\frac{\partial^2 z}{\partial x^2} - 3\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 56\cos(5x - 2y)$.	6	3	1

3. a) Solve the equation $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$, *given* $u(x,0) = 6e^{-3x}$ by separation of variables method. 6 4 2

b) Find the temperature in a bar of length *2* whose ends are kept at zero and lateral surface insulated if the initial temperature is $\sin\frac{\pi x}{2} + 3\sin\frac{5\pi x}{2}$. 6 4 2

4. a) Solve the equations
$27x + 6y - z = 85$; $x + y + 54z = 110$; $6x + 15y + 2z = 72$
by Gauss elimination method. 6 3 3

b) Apply factorization method to solve the system of equations $10x + y + z = 12$; $2x + 10y + z = 13$; $2x + 2y + 10z = 14$. (Take $u_{11} = 1$) 6 4 3

5. a) Using the Newton's divided difference formula , evaluate *f(8)* and *f(15)* given: 6 4 4

x	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

b) Given the following table of values of x and y 6 3 4

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y	1.000	1.025	1.049	1.072	1.095	1.118	1.148

Find $\frac{dy}{dx}$ *and* $\frac{d^2y}{dx^2}$ at x=1.05.

6. a) Find the value of y for $x=0.1$ and $x=0.2$ by Picard's method, given that $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$. 6 4 5

b) Apply Runge-Kutta method to find an approximate value of y for $x=1.1$, if $\frac{dy}{dx} = y^2 + xy$, $y(1) = 1$. 6 4 5

7. a) Find a root of the equation, using bisection method correct to three decimal places: $x^3 - x^2 - 1 = 0$. 6 3 1

b) Apply Euler's method to solve $y' = x + y$, $y(0) = 0$, choosing step length 0.2 (Carry out 6 steps) 6 3 5

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